

Exercise 1 Prove that the error growth of the differential equation

$$\frac{dx}{dt} = f(x)$$

is independent of the current state, i.e., $\delta x = \delta x(t)$, if and only if $f(x)$ is linear.

Exercise 2 Consider the Lorenz '63 model:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x), \\ \frac{dy}{dt} &= x(\rho - z) - y, \\ \frac{dz}{dt} &= xy - \beta z.\end{aligned}$$

Assume that $\sigma=10$, $\beta = \frac{8}{3}$, and $\rho=24.74$.

(a) Derive the linearized error evolution matrix $\mathbf{A}$ such that

$$\frac{d}{dt}(\delta \mathbf{x}) = \mathbf{A} \delta \mathbf{x}.$$

- (b) Compute the eigenvalues of $\mathbf{A}$ (using python). What is the physical significance of the eigenvalues of the linear operator $\mathbf{A}$ in this context?
- (c) Compute the determinant of $\mathbf{A}$ (using python). What is the meaning of $\det(\mathbf{A})$ here?